

About LLM Agent

This LLM Agent uses the Gemini model to reason about queries and take actions using available tools. It can segment objects in images by providing a textual query.

Available Tools

- Planning
- Web Search
- General Image segmentation
- Image Segmentation Evaluation
- One-shot Image Segmentation
- Mitochondrion Segmentation

Auto-Mode Settings

Max Segmentation Retries

GenCellAgent: Generalizable, Training-Free Cellular Image Segmentation

Upload your main image

image_13.png2.4x10

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.

Uploaded Main Image

Uploaded file path: /home/ldies/workspace/Storage/nyu1/persistent/GenCELLAgent-Copy1/examples/image_13.png

Select Segmentation Mode

Choose the appropriate mode for your task:

☒

 Auto Organelle Segmentation (ER, Golgi, Mito) - Uses General Text Guided Segmentation

☐

 Cell Segmentation - Uses specialized tools (CellSAM, Micro-SAM, Cellpose)

☐

 One-Shot Segmentation - Requires Example Image & Mask

You selected ☒ Auto Organelle Segmentation (ER, Golgi, Mito). No example images are required.

Enter a short query:

Help me segment the Golgi in the provide image.

Send

[Iteration 1] Agent Planning

Planning Steps:

```
{
  "0": {
    "step": 1
    "tool": "google"
    "condition": "No reference images provided and object characteristics not in knowledge base."
    "next_step_if_success": 2
    "next_step_if_failure": 2
  }
  "1": {
    "step": 2
    "tool": "imagesegmentation"
    "condition": "Perform initial segmentation using visual characteristics retrieved from a search."
    "next_step_if_success": 3
    "next_step_if_failure": 3
  }
  "2": {
    "step": 3
    "tool": "segmentationevaluation"
    "condition": "Evaluate the initial segmentation result to determine if it meets quality standards."
    "next_step_if_success": 6
    "next_step_if_failure": 4
  }
  "3": {
    "step": 4
    "tool": "imagesegmentation"
    "condition": "Re-run segmentation using the refined prompt provided by the evaluation to G1."
    "next_step_if_success": 5
    "next_step_if_failure": 5
  }
  "4": {
    "step": 5
    "tool": "segmentationevaluation"
    "condition": "Evaluate the refined segmentation result."
    "next_step_if_success": 6
    "next_step_if_failure": 4
  }
  "5": {
    "step": 6
    "tool": "summarizer"
    "condition": "Generate a final report of the segmentation process and results."
    "next_step_if_success": NULL
    "next_step_if_failure": NULL
  }
}
```

[Iteration 2] Agent Thought

Thought: I need to find the visual characteristics of the Golgi apparatus to create an effective segmentation prompt. I will search for its shape, texture, and typical location within a cell.

[Iteration 2] Agent Action

Action Details

- Name: google
- Reason: To retrieve visual characteristics of the Golgi apparatus for the initial segmentation prompt.

Input Parameters

```
{
  "search_query": "visual characteristics of Golgi apparatus in fluorescence microscopy shape texture location"
```

```

    "location": ""
  }
}

Action: Using google tool

Observation from google:

{
  "top_results": [
    {
      "position": 1
      "title": "Morphological determination of localization and function of ..."
      "link": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11103716/"
      "snippet": "The unique stacking and ribbon-like architecture of the Golgi apparatus forms the foundation for its precise functionality. Under cellular stress or ..."
    }
    {
      "position": 2
      "title": "How Do Proteins Move Through the Golgi Apparatus?"
      "link": "https://www.nature.com/scitable/topicpage/how-do-proteins-move-through-the-golgi-14397318/"
      "snippet": "In a stunning visual assay, both labs used live-cell fluorescence microscopy to directly observe cisternal maturation in Golgi of Saccharomyces cerevisiae ...."
    }
    {
      "position": 3
      "title": "Three-dimensional shape of the Golgi apparatus in different ..."
      "link": "https://academic.oup.com/jmicro/article-abstract/65/2/145/2579695"
      "snippet": "The full shape of the Golgi apparatus in epifidymal cells formed a basket-like structure with oval-shaped cisterns, while the Golgi apparatus in an acinar cell ..."
    }
    {
      "position": 4
      "title": "The Golgi Apparatus - Molecular Expressions Cell Biology"
      "link": "https://micro.magnet.fsu.edu/cells/golgi/golgiapparatus.html"
      "snippet": "Illustrated in Figure 2 is a fluorescence digital image taken through a microscope of the Golgi apparatus (pseudocolored red) in a typical animal cell. Note ..."
    }
    {
      "position": 5
      "title": "High-content phenotyping reveals Golgi dynamics and ..."
      "link": "https://rupress.org/jcb/article/225/1/e202503083/278467/High-content-phenotyping-reveals-Golgi-dynamics"
      "snippet": "Advances in cellular morphology analysis now enable the quantification of features, such as size, shape, and texture, providing insights into ..."
    }
  ]
}

summary: {
  visual_characteristics: "
  The visual characteristics of the Golgi apparatus across the provided sources focus on its unique structural complexity and dynamic nature. It is primarily described as a 'ribbon-like architecture' consisting of 'stacked, oval-shaped cisternae'. In three-dimensional space, its morphology varies by cell type, ranging from a 'basket-like structure' to localized clusters. In microscopy settings, it is typically visualized using 'live-cell fluorescence', often appearing as high-contrast, textured regions (frequently pseudocolored 'red') that exhibit distinct size and shape variations during cellular processes like cisternal maturation."
  segmentation_prompt: "
  Segment the Golgi apparatus characterized by its unique ribbon-like and stacked cisternal architecture. Identify the structure as a series of flattened, oval-shaped compartments or a basket-like formation, typically localized near the cell nucleus. Focus on regions exhibiting high-intensity fluorescence signals (often pseudocolored red in multi-channel images) with a distinct, granular texture. Differentiate the stacked ribbon layers from the surrounding cytoplasm, capturing both the localized clusters and the continuous filamentous connections between the stacks."
}
}
```

[Iteration 3] Agent Thought

Thought: I have retrieved the visual characteristics and a detailed segmentation prompt for the Golgi apparatus from the Google search. Now, I will perform the initial segmentation using the 'imagesegmentation' tool with the generated prompt.

[Iteration 3] Agent Action

Action Details

- Name: imagesegmentation
- Reason: To perform the initial segmentation of the Golgi apparatus using the visual characteristics retrieved from the search.

Input Parameters

```

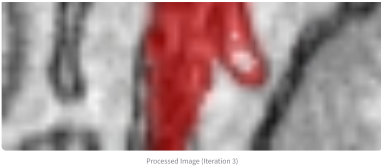
{
  "image_path": "
/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/examples/image_15.png"
  "prompt": "
Segment the Golgi apparatus characterized by its unique ribbon-like and stacked cisternal architecture. Identify the structure as a series of flattened, oval-shaped compartments or a basket-like formation, typically localized near the cell nucleus. Focus on regions exhibiting high-intensity fluorescence signals (often pseudocolored red in multi-channel images) with a distinct, granular texture. Differentiate the stacked ribbon layers from the surrounding cytoplasm, capturing both the localized clusters and the continuous filamentous connections between the stacks."
  "save_dir": "
/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/20260527_212654"
  "retry_count": 0
}
```

Action: Using imagesegmentation tool

Observation from imagesegmentation: Segmentation completed successfully in segment_save_path/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/20260527_212654/image_15_segmentation_v0.png, the corresponding mask saved in segment_mask_path/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/20260527_212654/image_15_segmentation_v0_mask.png

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.





[Iteration 4] 🧠 Agent Thought

Thought: I have completed the initial segmentation of the Golgi apparatus. Now, I need to evaluate the quality of this segmentation using the 'segmentationevaluation' tool to see if it meets the required standards or if a refined prompt is needed for a second attempt.

[Iteration 4] 🛠️ Agent Action

Action Details

- **Name:** segmentationevaluation
- **Reason:** To evaluate the initial segmentation result and obtain a score and a refined prompt if necessary.

Input Parameters

```
{
  "image_path":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/examples/imag
e_15.png"
  "mask_path":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images
/20260527_212654/image_15_segmentation_v0_mask.png"
  "segmentation_prompt":
    "Segment the Golgi apparatus characterized by its unique ribbon-like and stacke
d cisternal architecture. Identify the structure as a series of flattened, oval
-shaped compartments or a basket-like formation, typically localized near the c
ell nucleus. Focus on regions exhibiting high-intensity fluorescence signals (o
ften pseudocolored red in multi-channel images) with a distinct, granular textu
re. Differentiate the stacked ribbon layers from the surrounding cytoplasm, cap
turing both the localized clusters and the continuous filamentous connections b
etween the stacks."
  "save_dir":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images
/20260527_212654"
}
```

Action: Using segmentationevaluation tool

🔍 Observation from segmentationevaluation:

```
{
  "evaluation": {
    "quality_score": 0.33
    "criteria_scores": {
      "morphological_accuracy": 0.3
      "boundary_precision": 0.2
      "stack_organization": 0.1
      "completeness": 0.4
      "specificity": 0.8
      "structural_fidelity": 0.2
    }
  }
  "visual_guidance":
    "Golgi appears as a series of thin, parallel cisternae with distinct membra
ne boundaries. Detect 4-5 separate, elongated layers while maintaining the
narrow intercisternal gaps. Include small, spherical peripheral vesicles se
en flanking the central stack. Avoid merging the entire stack into a single
solid mass to preserve the characteristic stratified architecture."
  "overall_score": 33
  "refined_segmentation_prompt":
    "Segment the Golgi apparatus by identifying 4-5 distinct, thin, parallel cister
nae with elongated layers and clearly defined membrane boundaries. Preserve the
stratified architecture by maintaining narrow intercisternal gaps and avoiding
the merger of the stack into a single solid mass. Include small, spherical peri
pheral vesicles flanking the central stack while capturing the characteristic r
ibbon-like connections localized near the nucleus."
  "comparison_image_path":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images
/20260527_212654/image_15_comparison.png"
}
```

[Iteration 5] 🧠 Agent Thought

Thought: The initial segmentation score was low (0.33). I will now perform a second segmentation attempt using the refined prompt provided by the evaluation tool to improve the accuracy of the Golgi apparatus segmentation, specifically focusing on the stratified architecture and distinct cisternae.

[Iteration 5] 🛠️ Agent Action

Action Details

- **Name:** imagesegmentation
- **Reason:** To re-run the segmentation with the refined prompt to improve the quality score as per the evaluation feedback.

Input Parameters

```
{
  "image_path":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/examples/imag
e_15.png"
  "prompt":
    "Segment the Golgi apparatus by identifying 4-5 distinct, thin, parallel cister
nae with elongated layers and clearly defined membrane boundaries. Preserve the
stratified architecture by maintaining narrow intercisternal gaps and avoiding
the merger of the stack into a single solid mass. Include small, spherical peri
pheral vesicles flanking the central stack while capturing the characteristic r
ibbon-like connections localized near the nucleus."
  "save_dir":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images
/20260527_212654"
  "retry_count": 1
}
```

Action: Using imagesegmentation tool

🔍 Observation from imagesegmentation: Segmentation completed successfully in
segment_save_path/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-
Copy1/output_images/20260527_212654/image_15_segmentation_v1.png, the corresponding
mask saved in segment_mask_path/home/ides/workspace/Storage/xyul/persistent
/GenCELLAgent-Copy1/output_images/20260527_212654
/image_15_segmentation_v1_mask.png

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.





Processed Image (Iteration 5)

[Iteration 6] 🧠 Agent Thought

Thought: I have completed the second segmentation attempt (v1) using the refined prompt. Now, I must evaluate this result using the 'segmentationevaluation' tool to determine the final quality score and compare it with the previous attempt, as per the required workflow.

[Iteration 6] 🛠️ Agent Action

Action Details

- **Name:** segmentationevaluation
- **Reason:** To evaluate the second segmentation result (v1) and determine if the quality has improved.

Input Parameters

```
{
  "image_path":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/examples/image_15.png"
  "mask_path":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/28260527_212654/image_15_segmentation_v1_mask.png"
  "segmentation_prompt":
    "Segment the Golgi apparatus by identifying 4-5 distinct, thin, parallel cisternae with elongated layers and clearly defined membrane boundaries. Preserve the stratified architecture by maintaining narrow intercisternal gaps and avoiding the merger of the stack into a single solid mass. Include small, spherical peripheral vesicles flanking the central stack while capturing the characteristic ribbon-like connections localized near the nucleus."
  "save_dir":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/28260527_212654"
}
```

Action: Using segmentationevaluation tool

🔍 Observation from segmentationevaluation:

```
{
  "evaluation": {
    "quality_score": 0.27
    "criteria_scores": {
      "morphological_accuracy": 0.3
      "boundary_precision": 0.2
      "stack_organization": 0.1
      "completeness": 0.4
      "specificity": 0.4
      "structural_fidelity": 0.2
    }
  }
  "visual_guidance":
    "Detect individual, elongated cisternae as distinct parallel layers rather than a single merged mass. Maintain the narrow intercisternal spaces to preserve the characteristic stacked organization of the Golgi. Include peripheral dilated sacs and small associated vesicles that are currently omitted. Avoid merging separate membrane-bound compartments into a single blocky mass. Ensure segmentation boundaries tightly follow the dark membrane edges of each individual cisterna."
  }
  "overall_score": 27
  "refined_segmentation_prompt":
    "Segment the Golgi as distinct, elongated cisternae by tracing the dark membrane edges of each parallel layer individually. Maintain narrow intercisternal spaces to preserve the stacked organization, strictly avoiding the merger of separate compartments into a single blocky mass. Include peripheral dilated sacs and small associated vesicles while ensuring all segmentation boundaries tightly adhere to the membrane contours."
  "comparison_image_path":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/28260527_212654/image_15_comparison.png"
}
```

[Iteration 7] 🧠 Agent Thought

Thought: I have completed two segmentation attempts for the Golgi apparatus. The first attempt (v0) received a quality score of 0.33, and the second attempt (v1) received a score of 0.27. Since I have reached the maximum limit of two segmentation attempts as per the instructions, I will now proceed to generate a final report using the summarizer tool.

[Iteration 7] 🛠️ Agent Action

Action Details

- **Name:** summarizer
- **Reason:** To generate a final report of the segmentation process and results after completing the maximum allowed attempts.

Input Parameters

```
{
  "save_dir":
    "/home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/28260527_212654"
}
```

Action: Using summarizer tool

📄 Summarizer Report

View Full Report

Final Segmentation Report: Golgi Apparatus

1. Reference Detail Extraction

- **Original Image Path:** /home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/examples/image_15.png
- **Target Object:** Golgi Apparatus
- **Best Segmentation Mask Path:** /home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/28260527_212654/image_15_segmentation_v0_mask.png (Score: 0.33)
- **Secondary Segmentation Mask Path:** /home/ides/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/28260527_212654/image_15_segmentation_v1_mask.png (Score: 0.27)

2. Visual Characteristics of the Golgi Apparatus

Based on scientific literature and visual analysis, the Golgi apparatus in this image was identified by:

- **Architecture:** A complex, ribbon-like structure composed of 4-5 stacked, flattened, oval-shaped cisternae.
- **Location:** Typically localized in the perinuclear region (near the cell nucleus).
- **Texture/Signal:** High-intensity fluorescence with a distinct granular or filamentous texture.
- **Morphology:** Presence of small, spherical peripheral vesicles flanking the central stack and narrow intercisternal gaps between the parallel layers.

3. Process History & Tool Execution

The segmentation workflow followed a structured approach to maximize accuracy:

1. **Google Search** (🔍): Successfully retrieved detailed morphological data and generated an initial descriptive prompt.

2. Initial Segmentation (`imagesegmentation` - v0): Used the text-guided prompt to generate the first mask.

3. Evaluation 1 (`segmentationevaluation`): Assessed the v0 mask.

Score: 33.0/100

Feedback: Noted that while localization was good (Specificity 0.8), the stack organization was poor (0.1), suggesting the need to differentiate parallel cisternae rather than merging them.

4. Refined Segmentation (`imagesegmentation` - v1): Attempted a second segmentation using a prompt focused on preserving the stratified architecture and individual layers.

5. Evaluation 2 (`segmentationevaluation`): Assessed the v1 mask.

Score: 27.0/100

Feedback: The score decreased, primarily due to reduced specificity. The model struggled to maintain the narrow gaps between cisternae, resulting in a "blocky" mask.

4. Evaluation Outcomes & Summary

- Highest Quality Score: 33.0
- Performance Analysis: The automated segmentation successfully localized the Golgi region based on signal intensity. However, capturing the intricate, fine-scale gaps between individual cisternal layers remains a significant challenge for general text-guided segmentation.
- Comparison Image: A visual comparison between the original image and the segmentation masks is available at `/home/.idtes/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/20268527_212654/image_15_comparison.png`.

This report provides a transparent record of the segmentation attempts, highlighting the difficulty in resolving high-density organelle substructures without specialized training data.

[Iteration 8] @Agent Thought

Thought: The segmentation process for the Golgi apparatus has been completed. I have performed two segmentation attempts (v0 and v1) and evaluated both. The first attempt (v0) achieved a quality score of 0.33, while the second attempt (v1) achieved a score of 0.27. Following the evaluation of the second attempt, I invoked the summarizer tool, which has provided a comprehensive final report detailing the visual characteristics of the Golgi, the tool execution history, and the evaluation outcomes. Since the summarizer tool has already been used and its observation is available, I have sufficient information to provide the final answer and conclude the task.

Final Answer: The Golgi apparatus segmentation task is complete. The process involved retrieving visual characteristics via Google, performing two rounds of text-guided segmentation using the `imagesegmentation` tool, and evaluating each result.

Key Results:

- Best Segmentation:** The initial attempt (v0) was the most successful, achieving a quality score of 33.0/100.
- Best Mask Path:** `/home/.idtes/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/20268527_212654/image_15_segmentation_v0_mask.png`
- Visual Characteristics:** The Golgi was identified as a ribbon-like, stacked structure of flattened cisternae, typically located near the nucleus with high-intensity fluorescence.
- Challenges:** While the model successfully localized the organelle, resolving the fine-scale gaps between individual parallel layers proved difficult, leading to some merging of the cisternae in the masks.

A detailed comparison image showing the original image alongside the segmentation results can be found at `/home/.idtes/workspace/Storage/xyul/persistent/GenCELLAgent-Copy1/output_images/20268527_212654/image_15_comparison.png`.

HTL Summary: ## PART 1: Current Run Analysis

1. Segmentation Process: The current run followed a fully automated, text-guided segmentation workflow. The agent first performed an external search to define the morphological characteristics of the target (Golgi apparatus). It then executed two iterations of segmentation: an initial attempt based on the search results and a second attempt using a refined prompt generated by an automated evaluation tool. The process concluded with a summary report comparing the two versions.

2. Tools Used:

1. google: Used to retrieve scientific descriptions of the Golgi apparatus (ribbon-like, stacked cisternae) to construct a high-quality initial prompt.

2. imagesegmentation (v0): Performed the first text-to-mask generation based on the retrieved characteristics.

3. segmentationevaluation (v0): Provided an automated quality score (33.0) and specific visual guidance for refinement.

4. imagesegmentation (v1): Re-ran the segmentation using the refined prompt to attempt to improve structural fidelity.

5. segmentationevaluation (v1): Evaluated the second attempt, noting a decrease in score (27.0).

6. summarizer: Compiled the final report, documenting the workflow and outcomes.

3. User Interaction Behavior:

- Automatic vs. Manual:** The user chose a "Fully Automatic" mode from the outset. No manual correction tools (like `humanconnection`) were utilized.
- Use of References:** No reference images or masks were provided; the user did not utilize `onioncosegmentation`.
- Feedback Frequency:** The user provided no intermediate feedback; the "feedback" loop was entirely handled by the `segmentationevaluation` tool.
- Number of Iterations:** The workflow ran for the maximum of 2 automated iterations.

[CURRENT RUN] Recommended HTL Mode: Fully Automatic Reason: The user explicitly selected the "Auto Organelle Segmentation" mode and allowed the agent to handle knowledge retrieval, execution, and evaluation without any manual intervention or request for refinement.

PART 2: Long-Term User Profile and Final Recommendation

5. Behavioral Trends:

- Interaction Level:** This is the initial session recorded. Currently, the user shows a high reliance on the agent's autonomous capabilities.
- Tool Exploration:** The user is utilizing the "General Text Guided Segmentation" path, which leverages LLM-driven prompt engineering rather than visual examples.
- Shift/Progression:** As this is the first run, there is no established shift yet. However, the low quality scores (33.0 and 27.0) suggest that the user may eventually need to transition to "Reference Guided" or "Human Interaction" modes if the automated results do not meet their scientific standards.

6. User Profile: "The user currently prefers a fully automated, hands-off workflow. They rely on the agent to perform background research (via Google) and self-evaluate results. There is a clear preference for speed and automation over manual precision at this stage of their progression."

7. Final Recommendation:

[OVERALL RECOMMENDATION] Recommended HTL Mode: Fully Automatic User Profile: Initial entry shows a strong preference for fully automated workflows with minimal feedback. The user relies on text-guided prompts and automated evaluation rather than providing visual references or manual corrections. Reason: Based on the current session, the user chose the most automated path available. Until the user demonstrates a need for higher precision through manual tools or provides reference data, the Fully Automatic mode remains the most consistent fit for their established behavior.

Task Completed!

5 of 5

5/27/26, 21:30